

An exponential-size positive splitting in the reduced algebra

Walsh symmetrization and the full-activity obstruction

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Status. Complete proof candidate. The construction produces rational piecewise constant fair densities, not finite uniform dice: some cells have all letters active. The last proposition identifies a genuine obstruction to eliminating those cells within this particular family. No claim of literature novelty is made.

A multigraded cancellation lemma

Work in the associative algebra over $\mathbb{Q}$ generated by $H, Y_1, \dots, Y_n$, truncated above total degree n , and quotient further by every word in which some Y_i occurs twice. There is no restriction on repeated H . All exponentials and logarithms are finite formal sums. For $b \in \mathbb{F}_2^n$, let ρ_b fix H and send Y_i to $(-1)^{b_i} Y_i$.

Choose rows $b_1, \dots, b_L \in \mathbb{F}_2^n$. Starting with $W_0 = \exp(H + \sum_i Y_i)$, define

$$W_\ell = W_{\ell-1} \rho_{b_\ell}(W_{\ell-1}).$$

For $S \subseteq [n]$ put

$$w_\ell(S) = \#\{j \leq \ell : \sum_{i \in S} (b_j)_i = 1 \pmod{2}\}.$$

The Y -support of a nonzero monomial is the set of its Y_i letters; it contains each index at most once.

Lemma 1. *Every term of $\log W_\ell$ with nonempty Y -support S has total degree at least $w_\ell(S) + 1$. The terms with empty Y -support equal $2^\ell H$.*

Proof. At $\ell = 0$ the assertion is immediate. Write $A = \log W_{\ell-1}$ and expand $\log(e^A e^{\rho_{b_\ell} A})$ by the finite BCH formula. If S has odd parity under b_ℓ , its contribution from $A + \rho_{b_\ell} A$ cancels. Otherwise its old degree bound is precisely the required new bound.

Each remaining BCH term is a bracket of $k \geq 2$ homogeneous components with disjoint nonempty supports S_j , possibly together with components of empty support. Empty-support components have degree one. By induction the degree of the bracket is at least

$$k + \sum_j w_{\ell-1}(S_j) \geq k + w_{\ell-1}\left(\bigcup_j S_j\right),$$

since Hamming weight of a sum modulo two is at most the sum of the Hamming weights. This is at least the old bound plus one, which is sufficient whether the new parity is odd or even. Brackets containing only H vanish, proving the empty-support assertion. $\square$

Theorem 2. *For each $n \geq 1$ there is a positive piecewise constant path with $2^{O(n)}$ equal rational cells, each velocity of the form*

$$2 \sum_{i \in S} X_i \quad (S \subseteq [n]),$$

whose signature in the reduced free algebra is exactly $\exp(2^L \sum_i X_i)$. Equivalently, after normalization it is a permutation-fair family of n continuous probability distributions with densities taking only the values zero and two.

Proof. Choose an injective binary linear code $B : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^L$ with minimum nonzero weight at least n , where $L = 16n$ suffices. Indeed, take a random binary $L \times n$ matrix. For each nonzero input the output has the $\text{Bin}(L, 1/2)$ weight law; Chernoff's bound gives

$$\Pr(\text{wt}(Bs) < n) \leq e^{-49n/16}.$$

A union bound over fewer than 2^n nonzero inputs is less than one. Thus such a matrix exists, and its distance also ensures injectivity.

Use its rows in Lemma ???. Every nonempty support then has $w_L(S) \geq n$, so all its log terms vanish after truncation. Therefore $W_L = \exp(2^L H)$. Substitute

$$H = \sum_i X_i, \quad Y_i = X_i$$

in the reduced algebra on the original X_i . This substitution is well-defined: a repeated Y_i becomes a repeated original letter and vanishes. Each of the 2^L exponential factors has velocity $\sum_i (1 + \sigma_i) X_i$, where $\sigma_i \in \{-1, 1\}$. Finally scale each cell duration by 2^{-L} . The first-level coefficients are then one, and the full reduced signature is $\exp(\sum_i X_i)$, which is exactly uniformity of every distinct-letter order probability. $\square$

Why this does not yet finiteize by smaller alphabets

Proposition 3. *In the construction above, each of the 2^n activity subsets occurs exactly 2^{L-n} times. The assertion remains true after arbitrary fixed sign changes of the Y_i and after reordering the L binary symmetrizations. In particular, a full-activity cell cannot be eliminated by selecting an affine phase or by reordering those symmetrizations.*

Proof. The exponential factors are indexed by $t \in \mathbb{F}_2^L$, and their sign vector is $((-1)^{(B^T t)_i})_{i=1}^n$, in some binary order. The map B^T is onto because B has rank n . Every sign vector has exactly 2^{L-n} preimages. Fixed phases translate the sign vectors; row reordering merely permutes the domain coordinates. $\square$

The full-activity factor is a positive multiple of the uniform direction H . At an endpoint of an already fair path it can be cancelled while preserving fairness. Interior factors cannot in general be removed: $A \exp(aH) B = \exp(cH)$ does not imply $AB = \exp((c-a)H)$. Thus the present proof does not give a finite uniform-dice construction of exponential size. Finiteizing every cell by a fair set on its active alphabet is circular at the full-activity cells.

Proposition 4. *Suppose a family of nonconstant Walsh-character densities $f_i = 1 + \epsilon_i \chi_i$, with $\epsilon_i \in \{-1, 1\}$, has at most $n/2$ active coordinates on every equal cell. If $n \geq 4$, that family cannot be three-wise fair.*

Proof. Each nonconstant character has mean zero, so the mean number of active coordinates is $n/2$. The assumed upper bound is therefore equality on every cell, and $\sum_i \epsilon_i \chi_i = 0$ identically. Independence of distinct Walsh characters implies that each character occurs equally often with the two signs.

Write $G_\chi(t) = \int_0^t \chi(s) ds$. The distinct CDF profiles are $t + G_\chi(t)$ and $t - G_\chi(t)$. If two different character classes occur, their four distinct CDFs obey the nontrivial linear relation $(t + G_\chi) + (t - G_\chi) = (t + G_\psi) + (t - G_\psi)$. This contradicts the proved theorem that

distinct CDFs in a continuous three-wise fair family are linearly independent. If only one class occurs, each of its two distinct profiles occurs $n/2 \geq 2$ times, contradicting the companion statement that at most one distinct CDF profile can be repeated. Both inputs are proved in `two_dimensional_equalization_barrier.tex`. $\square$